

ARTICLE INFO

Keywords:

Solid oxide fuel cell
Marine power source
Ammonia
Anode off-gas recirculation
Ejector
Parametric study

ABSTRACT

Ammonia-fueled solid oxide fuel cell (SOFC) systems with anode off-gas recirculation (AOGR) offer high efficiency for marine power generation, but their performance is strongly coupled to operating conditions and the AOGR ratio. However, previous studies have generally treated AOGR ratio as a freely adjustable design input rather than a physically constrained outcome, and have examined its effect along a single operating axis. This study addresses these limitations by identifying the physically attainable maximum AOGR ratio and quantifying its interaction with operating conditions through a two-dimensional parametric study spanning fuel utilization, current density, and stack inlet temperature. The analysis was conducted using an integrated thermodynamic system model incorporating validated component models, implemented within a commercial thermodynamic simulator (H-Twin[®]). Results show that increasing fuel utilization from 40% to 80% raises the maximum AOGR ratio from 65.4% to 66.9%, whereas increasing current density from 0.1 to 0.5 A/cm² lowers it from 67.3% to 62.0%. Elevated stack inlet temperature further restricts this limit by degrading ejector entrainment performance. The AOGR ratio consistently improves efficiency while reducing power density, with the largest efficiency gains occurring at low fuel utilization. In addition, the optimal stack operating temperature shifts downward as the AOGR ratio increases. At high current density, growing balance-of-plant parasitic power produces a non-monotonic efficiency trend, yielding an optimal AOGR ratio below its maximum allowable value. These findings demonstrate that the AOGR ratio, operating conditions, and system constraints must be co-optimized for effective ammonia-fueled SOFC system design.

1. Introduction

The maritime industry is currently undergoing a major transition toward decarbonization, driven by tightening international regulations on greenhouse gas emissions and the long-term goal of achieving net-zero shipping [1, 2]. As conventional marine fuels such as heavy fuel oil and marine diesel oil are no longer compatible with these targets, the search for clean and energy-dense alternative fuels has become a central topic in marine power generation research [3]. Among the candidates proposed thus far, ammonia has emerged as one of the most prominent options because it contains no carbon and can therefore be utilized without any direct CO₂ emission [4]. Compared with hydrogen, which has often been regarded as the ultimate clean fuel but suffers from cryogenic storage requirements and low volumetric energy density, ammonia can be stored and transported as a pressurized liquid under significantly milder conditions,

making it far more practical for shipboard storage where space and weight constraints are critical [5, 6]. Furthermore, an extensive global infrastructure for ammonia production, transportation, and bunkering has already been established for fertilizer use, providing a uniquely favorable starting point for its near-term deployment as a marine fuel [7].

To convert ammonia fuel into electrical power onboard, a power conversion device that combines high efficiency with operational compatibility with ammonia is required. Internal combustion engines are an option currently under active development, but their combustion processes inherently produce NO_x and N₂O emissions that necessitate additional flue-gas treatment systems [8]. Fuel cells, in contrast, electrochemically convert fuel into electricity without combustion, thereby offering both higher efficiency and substantially cleaner emissions [9]. Among various fuel cell types investigated for ammonia operation, solid oxide fuel cell (SOFC) is particularly well-suited because its high operating

*Corresponding author.

 jongsup.hong@yonsei.ac.kr (J. Hong)

ORCID(s):

temperature, typically between 600 and 900 °C, simultaneously provides high thermodynamic efficiency and the thermal driving force necessary to decompose ammonia into hydrogen and nitrogen [10, 11]. For these reasons, ammonia-fueled SOFCs have rapidly emerged as a strong candidate for next-generation marine power generation systems pursuing zero-emission shipping.

Reflecting such potential, extensive research efforts have been conducted on ammonia-fueled SOFC systems, ranging from fundamental electrochemical performance evaluation to system-level thermodynamic analysis [12, 13, 14, 15, 16]. Wojcik et al. [12] experimentally evaluated ammonia as a fuel for SOFCs and demonstrated that ammonia-fed operation can achieve performance comparable to hydrogen-fed operation when ammonia decomposition is effectively promoted over suitable catalyst materials. Cinti et al. [13] combined ammonia-fed SOFC stack testing with system-level analysis and showed that SOFC operation with ammonia is technically feasible. Ishak et al. [14] performed thermodynamic analysis of ammonia-fed SOFCs and showed that ammonia can serve as an effective hydrogen carrier for SOFC operation, with system performance strongly affected by operating conditions such as stack temperature and fuel utilization. Lee et al. [15] thermodynamically compared internal and external ammonia decomposition strategies for ammonia-fueled SOFC systems, showing that thermal integration methods and system configurations can greatly affect system performance. Ishak et al. [16] conducted energy and exergy analyses of direct ammonia SOFC system integrated with a gas turbine cycle, showing that the hybrid configuration can effectively utilize SOFC waste heat for combined heat and power generation, improving overall system performance.

Previous studies have demonstrated that the performance of SOFC systems is strongly influenced by a wide range of design and operating parameters. A promising design strategy to enhance the electrical efficiency of SOFC systems is anode off-gas recirculation (AOGR). AOGR improves system electrical efficiency by decreasing the amount of fresh fuel consumption at a fixed SOFC stack fuel utilization condition, thereby increasing the net system fuel utilization [17]. To thoroughly analyze the effect of AOGR on ammonia-fueled SOFC systems, various studies have been conducted. Selvam et al. [18] thermodynamically analyzed a full AOGR ammonia-fueled SOFC system achieving 100% system fuel utilization, showing that the full AOGR strategy improves energy and exergy efficiencies by 12.17 and 11.39 percentage points, respectively, over a conventional system. Jeong et al, Bakała et al. [19, 20] compared the performance of SOFC systems fueled with hydrogen, natural gas and ammonia, under both with and without AOGR. Both studies revealed that AOGR generally tends to improve system electrical efficiency, regardless of fuel types. Quach et al. [21] studied the effect of water removal at the anode off-gas via condensation in ammonia-fueled SOFC system, and further improved system performance via integration of an internal combustion engine. The efficiency of AOGR system with water removal was 12.7 percentage point higher than that of a standalone system, and was only 0.1 percentage point lower than that of a AOGR system utilizing both water removal and internal combustion engine. Sun et al. [22] conducted a parametric study to analyze how operating conditions such as current density, AOGR ratio, etc. affects the system performance of an ammonia-fueled SOFC system. Oh et al. [23] also conducted a parametric study to find the optimal operating condition of an ammonia

fueled-SOFC system undergoing AOGR, integrated with an organic Rankine cycle.

However, while previous studies focus on proposing high-efficiency system architectures, they tend to lack practical thermodynamic considerations in system evaluation regarding AOGR. First, prior studies consider AOGR ratio as an independently adjustable operating variable, whereas in practice an upper bound may exist due to physical constraints such as pressure drops in BoPs [18, 21, 22, 23]. Second, the influence of AOGR ratio has typically been examined along a single operating axis at a fixed fuel utilization or current density, leaving the coupling between AOGR ratio and other operating variables unexplored [21, 22, 23]. Third, prior studies either do not account for or assume a designated value for pressure drop effects at heat exchangers [18, 20, 21, 22, 23]. Furthermore, in systems utilizing an ejector for AOGR, the ejector model is overly simplified as a component which mixes two streams and pressurizes the entrained flow by a fixed pressure ratio [22].

To address such limitations, the present study pursues the following research objectives with the following novelties:

- The physically attainable maximum AOGR ratio is identified as an endogenous outcome of component level hydrodynamics under varying stack inlet temperature, fuel utilization, and current density, rather than being prescribed as an external design input.
- A two-dimensional parametric study is conducted in a system level to reveal how the AOGR ratio couples with each operating variables, thereby analyzing the mechanisms by which the AOGR ratio influences overall system performance under realistic operating conditions.

- A high-fidelity ammonia-fueled SOFC system model is developed, in which pressure drops across balance-of-plant components are evaluated through empirical correlation based physical models accounting for design variables, and the AOGR is described by a reliable one-dimensional ejector model that captures the entrainment-pressure-lift behavior.

2. Thermodynamic Component Modeling

Methodology

To study the performance of ammonia-fueled SOFC systems under various operating conditions, thermodynamic systems were computationally modelled and analyzed via numerical simulation. Numerical simulations were conducted via a commercial solid oxide cell system simulator, H-Twin® (H-Cube Solutions). The thermodynamic systems consisted of system components, fluid streams, and an integrated system. An integrated system was assembled by a set of system components which are connected by fluid streams represented by thermodynamic properties including mass flow rate ($\dot{m}$), temperature (T), pressure (p) and chemical composition (y_i). The system components of an ammonia-fueled SOFC system include SOFC stack, heat exchanger, ammonia reformer, combustor, blower, etc. In addition, if the system undergoes AOGR, an ejector is also included to induce suction of the relatively low-pressure anode off-gas. Thermodynamic properties are calculated for each stream via a steady-state numerical solver, where each system component calculates the thermodynamic properties of its outlet stream as the output, given the inlet stream and its design configurations as the input. Numerical values of thermodynamic properties were acquired from the following sources: thermodynamic properties of water [24], specific heat capacities of gases [25], thermal conductivity and

dynamic viscosity of gases [26], and enthalpy of formation of gases [27]. Design configuration of system components include geometrical design variables and operational parameters. For example, in the case of SOFC stack, geometrical design variables such as cell dimensions and operational parameters such as current density are determined as design configurations. Together with the design configurations and thermodynamic properties of its cathode/anode inlet stream as the input, the thermodynamic properties of its outlet streams and voltage, power generation values are calculated as the output by solving internal electrochemistry and mass/energy balance equations. Throughout the study, key assumptions applied are as follows:

- For chemical species, only H_2 , H_2O , O_2 , N_2 , NH_3 are considered.
- For the equation of state, only H_2O is modeled as a real gas while other gases are modeled as ideal gases.
- In energy conservation, kinetic and potential energy are neglected.
- All system components assume an adiabatic condition between the component and the environment.
- Radiative heat transfer is neglected.
- Steady-state operation is assumed.
- Pressure drop does not occur in pipes when connecting components.
- Effect of stacking SOFC cells is neglected.

2.1. SOFC Stack

A 1 – dimensional model of SOFC stack developed by Lee [28] was exploited to calculate the voltage, power generation and thermodynamic properties of its outlet streams. The model considers discretized sections of a SOFC stack

in the direction of fuel flow and solves complete analytical expressions of electrochemical, thermodynamic relations for each section to derive its output parameters. Given the input parameters, internal electrochemical reaction is solved and the local value of voltage of each section is calculated by the following equations.

$$V = E_{rev} - \eta_{act,a} - \eta_{act,c} - \eta_{ohm} - \eta_{conc,a} - \eta_{conc,c} \quad (1)$$

$$E_{rev} = -\frac{\Delta G^\circ}{2F} + \frac{RT}{2F} \ln \left(\frac{p_{H_2} p_{O_2}^{\frac{1}{2}}}{p_{H_2O}} \right) \quad (2)$$

Detailed equations and parameters of overpotentials are provided in [28]. The model was validated against experimental data for two fuel conditions: pure ammonia and a 1:3 N_2/H_2 mixture representing fully decomposed ammonia. The calculated values of voltage at each section are averaged to derive the final voltage of the stack. Based on the value of voltage, design configurations and operational parameters of the stack, gross power generation ($\dot{W}_{gross}$) is calculated by the following equation,

$$\dot{W}_{gross} = j A_{cell} N_{cell} V_{SOFC} \quad (3)$$

where j is the current density, A_{cell} is cell effective area, N_{cell} is the number of stacked cells, V_{SOFC} is the unit cell voltage. Along with the calculation of voltage, the heat generated from electrochemical reaction is utilized for outlet gases post – heating, satisfying the mass, energy and heat balance in each section and the overall stack.

2.2. Balance of Plants (BoPs)

2.2.1. Heat Exchanger

A shell and tube heat exchanger model developed by Park et al. [29] was employed to simulate heat transfer from post – combustion gas to fresh air. Fresh air flows into the tube side of the heat exchanger as cold fluid, and post

combustion gas enters the shell side as the hot fluid. The model takes geometrical parameters such as tube length, shell diameter, number of baffles and thermodynamic properties of inlet gases as the input to calculate output values such as pressure drops, heat transfer coefficient, outlet gas temperature. Such output parameters are calculated from input parameters by the Bell – Delaware method.

2.2.2. Reformer

An ammonia reformer model developed by Lee et al. [15] was exploited to simulate ammonia decomposition using heat from the high temperature stack off-gas. The reformer model shares the same dimensional properties as the shell and tube heat exchanger model while the tube is packed with cylindrical pellets of Ni catalyst. Pure ammonia, or a mixture of ammonia and anode off-gas flows into the tube side as the cold fluid and stack off-gas enters the shell side as the hot fluid. It is assumed that the composition of outlet gas of the tube side reaches chemical equilibrium, as ammonia is fully decomposed above 600 °C in Ni-based catalysts[30, 31, 32].

2.2.3. Ejector

An ejector model developed by Park et al. [33] was employed for AOGR. The model has been validated in the same study by comparing simulation results with data from a prior study accounting ejector performance in a SOFC AOGR system [34]. Performance parameters of the validated ejector model are provided in [Supplementary Table S1](#). The ejector allows AOGR with zero parasitic power by ‘inhaling’ the low-pressure anode – off gas as entrained flow, using high – pressure ammonia as primary flow. As a result, the ejector discharges a mixture fluid as a fuel for SOFC stack. The ejector model accounts for compressible flow effects by assuming critical mode operation, where both the primary and secondary stream undergo choking. Input parameters

are the thermodynamic properties of both flows entering the ejector, and the model calculates a suitable geometry for mass flow rates of the inlet flows as the output, together with the thermodynamic properties of the mixed stream.

2.2.4. Others

Other BoPs composing the system include blower, splitter, combustor, etc. Under a similar underlying mechanism, the BoPs calculate the thermodynamic properties of its outlet stream by calculating mass – energy balance relations together with governing equations depending on the function of the BoP. Details of each model can be found in reference [29].

3. System Configuration

3.1. System Layout Design

When designing a SOFC system, a suitable system integration method must be adopted to satisfy thermal conditions such as the feasibility to supply sufficient heat to SOFC inlet flows. Accordingly, two system layouts were evaluated in this study: configurations with and without AOGR. The process flow diagrams of both layouts are presented in Fig. 1.

3.1.1. Recirculation System

The process flow diagram of the recirculation system layout is shown in Fig. 1a. This layout was obtained by referencing Ejector CG layout proposed by Park [33] as a baseline and modifying it to meet the objectives of this study. The three limitations of the baseline layout under the operating conditions considered in this study, together with the resolution applied to each of them, are described in the remainder of this section.

Ejector CG is a layout where a portion of anode off-gas is recirculated as fuel, and the rest is utilized as a hot stream of the reformer. Air and fuel are pressurized by a blower

of utilizing cathode off-gas as the heat source of the fuel reformer.

To resolve limitation 2, the heat duty value at both the heat exchanger and reformer must be adjusted to provide a suitable amount of heat to the cold side fluid. However, the heat duty required at each point differs for every different operating condition of the system. Since the heat exchanger model and reformer model calculates heat duty and outlet temperature based on design configuration and inlet thermodynamic properties, heat duty can be adjusted by differing the heat exchanger's geometrical parameters or differing the mass flow rate entering the shell side. Mass flowrate of the shell side fluid greatly affects the amount of heat transfer between the tube and shell side fluid, as a greater mass flowrate substantially increases turbulence in the shell side flow, improving heat transfer performance [35]. The design configurations of a shell and tube heat exchanger is a unique parameter which cannot be adjusted flexibly in real engineering applications; therefore, shell side inlet mass flowrate is a readily available variable that can be varied. Shell side inlet mass flowrate can be adjusted by adding a splitter upstream of the heat exchanger and reformer's shell side flow. The splitter is given a designated split ratio to branch off a suitable mass flowrate of hot gas to the heat exchanger and reformer, so that the cold outlet stream can reach a desired temperature. In the case of heat exchanger, mass flow rate is branched off to provide a suitable amount of heat for air heating, and in the case of reformer, mass flow rate is branched off to provide a suitable amount of heat for ammonia decomposition and fuel heating.

Limitation 3 occurs due to a difference in the target environment and fuel of the system. The study conducted by Park [33] analyzed a natural gas fueled SOFC system for residential heat and power generation, thereby assuming

that natural gas is fed into the system at 1 atm. To induce proper operation at the ejector and compensate for pressure drop effects at the fuel reformer, natural gas is pressurized by the compressor at a high pressure of over 5 atm in the corresponding study. Such operation causes a significant increase in parasitic power due to power consumption of the fuel compressor. However, this study considers ammonia as a fuel for marine power generation, where ammonia is conventionally stored at a high pressure of 18 bar for fuel [36]. Therefore, given the sufficiently high inlet pressure of pure ammonia, the fuel compressor is omitted, and a throttling valve is installed instead to meet the required pressure boundary conditions across operating points.

3.1.2. Non-Recirculation System

In a case where the AOGR ratio is 0, the ejector does not play any role in the system as no part of the anode off-gas is fed back, and pure ammonia fuel from the system inlet bypasses through the ejector. To evaluate non recirculation system performances under unified conditions, an identical system layout was utilized, but the ejector was omitted from the process flow diagram together with the recirculating splitter. Since the splitter is omitted, all parts of the anode off-gas is directly fed to the combustor to provide heat to the air heat exchanger after combustion. The process flow diagram of the non – recirculation system layout is shown in Fig. 1b.

3.2. Input Parameters & Boundary Conditions

To compare system performances at different operating conditions, a set of common input parameters must be established to solely analyze the effect of changing operating parameters. The input parameters and boundary conditions imposed in this study are summarized in Table 1. The ambient conditions are set as 25°C and 1 atm, thereby setting the

inlet condition of air as the same temperature and pressure. Due to the ambient condition, a pressure boundary condition is set at the system sink to be 1 atm, so that neither fluid backflow occurs at the sink nor any unnecessary parasitic power is consumed at the air blower. In addition, system outlet temperature is set to dew point, as a waste heat recovery system recuperates heat from the exhaust gas until it reaches dew point. At the fuel inlet side, 100% ammonia is fed as a saturated vapor at 18 bar, which is the pressure where fully pressurized ammonia is transported and stored as a marine fuel [36]. The temperature of system inlet ammonia is set to 45°C, which is the saturation temperature of ammonia at 18 bar. The pressure difference between anode and cathode inlet at the SOFC stack is set to be under 10 mbar to avoid mechanical destruction of the cell [37]. The temperature of the stack inlet stream is set to be minimum at 600°C among the parametric study to guarantee complete decomposition of ammonia at the fuel reformer and stack [32]. In addition, the temperature difference between the inlet stream and outlet stream of SOFC stack is set as 100°C to reduce thermal stress to the stack [38]. The mass flowrate of air through cathode inlet was adjusted at each operating conditions to satisfy the temperature difference boundary condition, as air can be utilized as a coolant to avoid overheating inside the stack. As fluid flows across the stack, it is assumed that a pressure drop of 0.5% occurs due to the geometry of anode and cathode channels [15].

The operating conditions of the system analyzed in this system are varied in a wide range with a base case stack inlet temperature (T_{stack}) of 600 °C, stack fuel utilization rate 70% (FU_{stack}), current density 0.2 A/cm² (j), AOGR ratio 50% (r), which is a condition to generate approximately 50 kW of gross power. Stack fuel utilization and AOGR ratio are defined as Eqs. (4) and (5).

Table 1. Input parameters and boundary conditions throughout this study.

Parameter	Value	Unit	Ref.
System boundary conditions			
System inlet air temperature	25	[°C]	
System inlet air pressure	1	[atm]	
Chemical composition of air	21% O ₂ , 79% N ₂	[%]	
System inlet fuel temperature	45	[°C]	[36]
System inlet fuel pressure	18	[atm]	[36]
Chemical composition of fuel	100% NH ₃	[%]	
System sink pressure	1	[atm]	
System sink temperature	Dew point	[°C]	
SOFC stack			
Number of stacked cells	2380	[ea]	
Effective cell area	121	[cm ²]	
Anode pressure drop	0.5	[%]	[15]
Cathode pressure drop	0.5	[%]	[15]
SOFC inlet pressure difference tolerance	10	[mbar]	[37]
SOFC inlet/outlet temperature difference	100	[°C]	[38]
Blower			
Isentropic efficiency	85	[%]	

$$FU_{stack} = \frac{\dot{n}_{O^{2-}}}{\left(\dot{n}_{H_2} + \frac{3}{2}\dot{n}_{NH_3}\right)_{stack,in}} \quad (4)$$

$$r = \frac{\dot{n}_{AOGR}}{\dot{n}_{stack,a,out}} \quad (5)$$

Where $\dot{n}_{O^{2-}}$ is the rate of oxygen ion transport, defined as Eq. (6).

$$\dot{n}_{O^{2-}} = \frac{jN_{cell}A_{cell}}{2F} \quad (6)$$

To meet the power demand under given operating conditions, the SOFC stack is consisted of 2380 cells with an active area of 121 cm². Along with the scale of SOFC stack, mass flowrates of fuel inlet at each operating condition is determined as Eq. (7).

$$\dot{m}_{NH_3} = \frac{2\dot{n}_{O_2} - MW_{NH_3}}{3FU_{stack}} \cdot (1 - r(1 - FU_{stack})) \quad (7)$$

Where MW_{NH_3} is the molar mass of ammonia. When AOGR is applied, unused fuel from the SOFC stack outlet is recirculated, and the recirculated fuel supplies the fuel demand at the anode inlet. Such process reduces the amount of fuel required at the system inlet while satisfying FU_{stack} conditions; hence the system fuel utilization is determined as Eq. (8).

$$FU_{system} = \frac{FU_{stack}}{1 - r(1 - FU_{stack})} \quad (8)$$

Dimensional parameters for the heat exchanger and reformer were determined following standard design guidelines for commercial shell-and-tube heat exchangers [39, 40, 41]. Key geometric variables—including tube length, shell diameter, and baffle spacing—were tuned to ensure that the required heat duties are met over a wide operating range. To prevent overdesign, the final dimensions were selected such that the tube-side velocity remained within an appropriate range (15-30m/s) for effective heat transfer [42]. However, since fluid flow rates vary substantially across operating conditions, a single heat exchanger design cannot satisfy the heat duty and velocity requirements at every point, even with splitter fluid control. Therefore, if a given heat exchanger unit could not physically meet its heat duty requirements or tube-side velocity range, its size would be adjusted to meet such conditions. Details about the adjustment of heat exchanger and reformer dimensions is described in [Supplementary Note S1](#). The main geometric parameters of the base-case heat exchanger and reformer are summarized in Table 2.

Table 2. Heat exchanger & reformer design configurations for base case operation

Parameter	Value	Unit	Ref.
Air heat exchanger			
Shell diameter	0.643	[m]	[39]
Tube diameter	0.0127	[m]	
Number of tubes	945	[ea]	
Baffle spacing	0.214	[m]	
Ammonia reformer			
Shell diameter	0.308	[m]	[39]
Tube diameter	0.0127	[m]	
Number of tubes	134	[ea]	
Baffle spacing	0.103	[m]	
Catalyst pellet diameter	0.002974	[m]	
Catalyst pellet length	0.01041	[m]	

Table 3. Variations in the operating parameters.

Operating parameters	Values	Unit
T_{stack}	600, 650, 700, 750, 800	[°C]
FU_{stack}	40, 50, 60, 70, 80	[%]
j	0.1, 0.2, 0.3, 0.4, 0.5	[A/cm ²]
AOGR ratio	0, 10, 20, 30, 40, 50... Max	[%]

3.3. Operating parameters for parametric study

To analyze the operating limits and performance effects of AOGR ratio via parametric study, the following operating parameters were varied together with AOGR ratio: T_{stack} , FU_{stack} , j . The range and values of parameter variation are described in Table 3.

4. Results and Discussion

To analyze the sole effect of each operating parameter, each parameters were primarily varied at the base case. The complete stream table for the base case operation is provided in [Supplementary Table S2](#). Key performance indices include system gross power, electrical efficiency (η_{elec}), etc. In this study, the only parasitic power induced in the system is power consumption at the air blower ($\dot{W}_{blower}$). Therefore, electrical efficiency and net power ($\dot{W}_{net}$) can be defined as Eqs. (9) and (10). To evaluate the performance of the

Table 4. Simulated performance indices of the base case operation.

Parameter	Value	Unit
Cell voltage	0.864	[V]
Gross power	49.74	[kW]
Power density	0.173	[W/cm ²]
Fuel thermal input (LHV)	76.54	[kW]
Air blower power	1.35	[kW]
Net power	48.40	[kW]
System fuel utilization	82.4	[%]
Electrical efficiency	63.23	[%]

system's capability to generate electrical power at a non-biased scale, the gross power was normalized as power density ($\dot{w}_{gross}$), defined as Eq. (11).

$$\eta_{elec} = \frac{\dot{W}_{net}}{\dot{n}_{fuel} LHV_{fuel}} \times 100, \quad (9)$$

$$\dot{W}_{net} = \dot{W}_{gross} - \dot{W}_{blower}, \quad (10)$$

$$\dot{w}_{gross} = \frac{\dot{W}_{gross}}{N_{cell} A_{cell}}. \quad (11)$$

4.1. Base-case system performance

Prior to the parametric investigation, simulation results of the base case (600 °C T_{stack} , 70% FU_{stack} , 0.2 A/cm² j , 50% r) are examined to establish a quantitative reference against which the trends discussed in the subsequent sections can be interpreted. The performance indices obtained at the base case are summarized in Table 4. The stack operates at a cell voltage of 0.864 V, which yields a gross power of 49.74 kW and a power density of 0.173 W/cm². Against a fuel thermal input of 76.54 kW on an LHV basis, the system consumes 1.35 kW of parasitic power at the air blower, resulting in a net power of 48.40 kW and an electrical efficiency of 63.23%. Since half of the unreacted fuel leaving the anode is returned to the ejector, the FU_{system} defined in Eq. (8) reaches 82.4%, appreciably higher than the 70% imposed at the stack.

4.2. Determination of maximum AOGR Ratio

The base case examined above recovers half of the unreacted anode off-gas and thereby attains a system fuel utilization of 82.4%, and Eq. (8) shows that greater r attains greater FU_{system} . Such trend raises the question of how far the AOGR ratio can actually be increased, as greater r and FU_{system} can potentially improve system efficiency by reducing fuel consumption. In the present layout the AOGR ratio cannot be prescribed at will, because the ejector operates without external power input and the only source available to return the recirculated stream to the anode inlet pressure is the stagnation pressure of the primary ammonia flow, which is bounded by the 18 bar storage condition. Once the pressure lift that the ejector can deliver becomes insufficient to overcome the pressure drops accumulated along the recirculation path, the pressure balance of the AOGR loop can no longer be closed and the corresponding operating becomes infeasible. The AOGR ratio is thus bounded by a feasibility limit rather than by a performance penalty, and this limit is not a fixed value, since both the required and the attainable pressure lift vary with FU_{stack} , j and T_{stack} . Determining this limit as a function of the operating condition is accordingly a prerequisite for the parametric analysis of Section 4.3, in which the AOGR ratio is swept up to its maximum allowable value at every operating point.

To inspect in detail the mechanism of determination of maximum AOGR ratio, the streams of the anode side gas and its recirculated portion can be detached from the overall system to compose an AOGR loop section as Fig. 2.

As a mixture of system inlet fuel and the recirculated portion of anode off - gas (stream 2) flows consecutively through the fuel reformer and stack anode channel, the stream undergoes pressure drop and exits the stack (stream 4). A portion of the low-pressure anode off-gas is branched

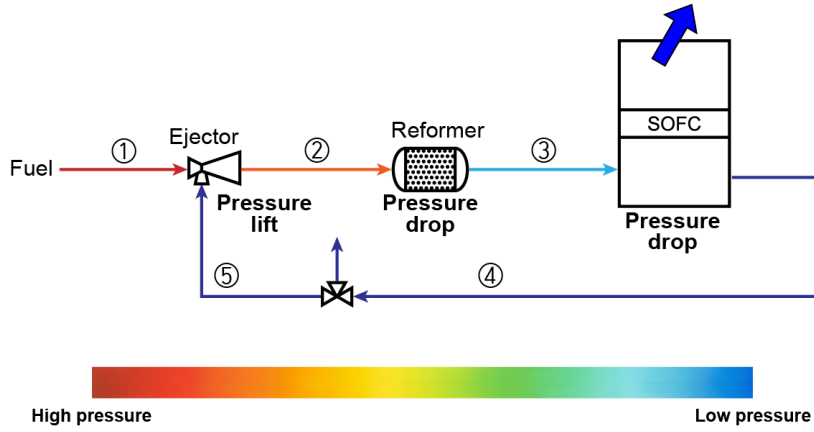


Figure 2: Schematic diagram of the AOGR loop section and pressure distribution in the constituent streams.

off through the splitter depending on the AOGR ratio (stream 5) and enters the suction inlet of the ejector. The pressure of stream 5 must be elevated to compensate for pressure drop effects of the fuel reformer and stack anode channel. Therefore, a high-pressure fuel (stream 1) enters through the primary inlet of the ejector and is mixed with stream 5 to pressurize the recirculated fluid. The mixed fluid leaves the ejector at a back pressure as stream 2, and the pressure balance of AOGR loop is satisfied.

As the system AOGR ratio increases, the entrainment ratio of the ejector also increases. Under this condition, a larger mass flow rate of the low-pressure fluid in stream 5 is entrained into the ejector, and therefore the pressure of stream 1 must be increased to ensure proper mixing with stream 5 and to achieve the desired back pressure at stream 2. This is because a higher primary inlet pressure increases the stagnation pressure of the primary flow, thereby supplying more kinetic energy to the mixed fluid and resulting in greater pressure recovery [43]. However, the maximum pressure of stream 1 is limited to 18 bar, and the 18 bar primary flow may not be able to sufficiently pressurize the recirculated stream when the AOGR ratio is too high. In addition to this, the pressure drop occurring at the fuel

reformer cold side also generally increases with the AOGR ratio due to an increase in the mass flowrate of stream 2. Therefore, it can be deduced that the maximum AOGR ratio of the system is determined by the following two factors: entrainment and pressurization performance of the ejector, and pressure drop occurring at the fuel reformer cold side.

Changes in operating conditions of the SOFC system (FU_{stack}, j, T_{stack}) can affect both the performance of the ejector and fuel reformer cold side pressure drop. Therefore, it is necessary to analyze how a change in each operating condition affects the maximum AOGR ratio permitted to the system.

In the case of FU_{stack} , its value directly determines the ammonia fuel flow rate supplied to the system inlet, thereby strongly affecting the mass flow rate of anode-side gases. As FU_{stack} increases from 40% to 80%, the ammonia fuel flow rate required to sustain the target electrochemical operating condition of the SOFC stack decreases, which in turn reduces the mass flow rate of the anode-side gases passing through the fuel reformer cold side. For a given packed bed reformer geometry, this reduction in flow rate diminishes the viscous friction effect within the catalyst bed, leading to a decrease in the cold side pressure drop. As shown in

Fig. 3(a), when FU_{stack} increases from 40% to 80%, the reformer cold side pressure drop decreases monotonically from 8.76 kPa at 40% FU_{stack} to 6.66 kPa at 80% FU_{stack} . Since a lower reformer cold-side pressure drop alleviates the discharge back pressure of the ejector imposed in the AOGR loop, the maximum allowable AOGR ratio that the system can sustain correspondingly increases from 65.4% to 66.9% over the same FU_{stack} range. This trend indicates that operating the system at higher FU_{stack} is operationally favorable for expanding the feasible AOGR operating window, as the reduced anode-side flow rate mitigates the parasitic pressure losses across the reformer and thereby relaxes the pressure-balance constraint that governs the upper bound of the AOGR ratio.

In the case of j , its value directly determines the amount of fuel electrochemically consumed in the SOFC stack, and thereby simultaneously increases both the air and ammonia fuel flow rates required at the system inlet. As j increases from 0.1 A/cm² to 0.5 A/cm², the ammonia fuel flow rate required to sustain the elevated electrochemical reaction rate rises accordingly, which in turn increases the mass flow rate of the anode-side gases passing through the fuel reformer cold side. Furthermore, a higher j leads to a larger heat duty for the endothermic ammonia decomposition reaction, which requires a longer fuel reformer length to provide sufficient heat transfer area. The combined effect of the increased flow rate and the extended reformer length amplifies the viscous friction loss along the packed catalyst bed, leading to a substantial increase in the cold side pressure drop. As shown in Fig. 3(b), when j increases from 0.1 A/cm² to 0.5 A/cm², the reformer cold side pressure drop increases monotonically from 6.50 kPa at 0.1 A/cm² to 11.14 kPa at 0.5 A/cm². Since a higher reformer cold-side pressure drop intensifies the discharge back pressure of the ejector imposed in the

AOGR loop, the maximum allowable AOGR ratio that the system can sustain correspondingly decreases from 67.3% to 62.0% over the same j range. This trend indicates that operating the system at higher j narrows the feasible AOGR operating window, as the simultaneous increase in anode-side flow rate and reformer length aggravates the parasitic pressure losses across the reformer and thereby tightens the pressure-balance constraint that governs the upper bound of the AOGR ratio.

In the case of T_{stack} , its value directly determines the temperature of the entrained anode off-gas returning to the ejector as well as the temperature of the gas entering the fuel reformer cold side. As T_{stack} increases from 600 °C to 800 °C, the elevated gas temperature reduces the gas density at a fixed mass flow rate, which in turn increases the volumetric flow velocity through the fuel reformer cold side. This higher flow velocity enhances the viscous friction loss along the packed catalyst bed, leading to a gradual increase in the cold side pressure drop. As shown in Fig. 3(c), when T_{stack} increases from 600 °C to 800 °C, the reformer cold side pressure drop increases monotonically from 6.96 kPa at 600 °C to 8.66 kPa at 800 °C. However, in contrast to the cases of FU_{stack} and j , the variation in the cold side pressure drop alone is insufficient to account for the magnitude of the change in the maximum allowable AOGR ratio, which decreases from 66.6% at 600 °C to 61.2% at 800 °C over the same temperature range. Such substantial reduction implies that, under varying T_{stack} , the upper bound of the AOGR ratio is governed not only by the reformer pressure drop but more dominantly by the entrainment performance of the ejector itself.

Specifically, as the temperature of the entrained anode off-gas rises while the primary flow condition remains fixed, the speed of sound of the secondary stream also increases,

which leads to a reduction in the convective Mach number that characterizes the compressible mixing process between the primary and secondary flows within the ejector. According to the recent numerical investigation of supersonic ejector mixing by Liang et al. [44] a lower convective Mach number—corresponding to a higher total temperature ratio between the secondary and primary flows—weakens the momentum exchange across the confined mixing layer, thereby deteriorating the mixing process between the two streams. Since insufficient mixing prevents the secondary flow from being adequately accelerated by the high-velocity primary flow, the Mach number of the mixed flow at the diffuser inlet decreases, which in turn degrades both the entrainment capability and the pressure lift performance of the ejector. Consequently, the ejector becomes less capable of overcoming the discharge back pressure of the AOGR loop, and the maximum allowable AOGR ratio decreases more sharply than what the reformer pressure drop alone would predict. To verify this, a supplementary analysis was conducted by modelling a simple two inlet-ejector-single sink system inhaling pure ammonia as primary flow and arbitrary anode off-gas as the entrained flow. Temperature of the entrained flow was varied from 700 °C to 900 °C, and the mach number and back pressure of the ejector exit stream was deduced. The results confirm that for a constant entrainment ratio, pressure lift performance of the ejector decreases as the temperature of entrainment flow increases (Supplementary Tables S3 and S4). This trend indicates that operating the system at higher T_{stack} is operationally unfavorable for sustaining a wide AOGR operating window, as the deteriorated performance becomes the dominant constraint that governs the upper bound of the AOGR ratio.

4.3. Effect of AOGR ratio at different operating conditions

While the previous section identified the maximum allowable AOGR ratio under each operating condition, the AOGR ratio itself is a key design variable that strongly influences the overall system performance within the feasible operating window. In this section, the effect of the AOGR ratio on the system performance is investigated under varying operating conditions, namely $FU_{stack,j}$, and T_{stack} . For each operating condition, the AOGR ratio is varied from zero up to the maximum allowable value determined in Section 4.2, and the resulting trends in system efficiency and key state variables are analyzed. This two-dimensional parametric analysis enables a comprehensive understanding of how the AOGR ratio interacts with each operating condition to shape the overall performance characteristics of the NH₃-fueled SOFC system.

4.3.1. Effect of AOGR ratio at different fuel utilizations

To investigate the influence of the AOGR ratio on the system performance under varying FU_{stack} , the AOGR ratio was swept from 0% up to the maximum allowable value identified in Section 4.2, while FU_{stack} was varied from 40% to 80%. The resulting trends in system efficiency and stack power density are summarized in Fig. 4(a), where each iso-fuel-utilization line traces the trajectory of the system operating point as the AOGR ratio increases.

As shown in Fig. 4(a), increasing the AOGR ratio consistently improves system efficiency while reducing stack power density across all FU_{stack} conditions. However, these two opposing effects respond differently to FU_{stack} : the efficiency gain varies significantly with FU_{stack} , whereas the power density penalty remains nearly insensitive to it. At a low FU_{stack} of 40%, raising the AOGR ratio from 0% to its maximum allowable value of 65.4% increases the

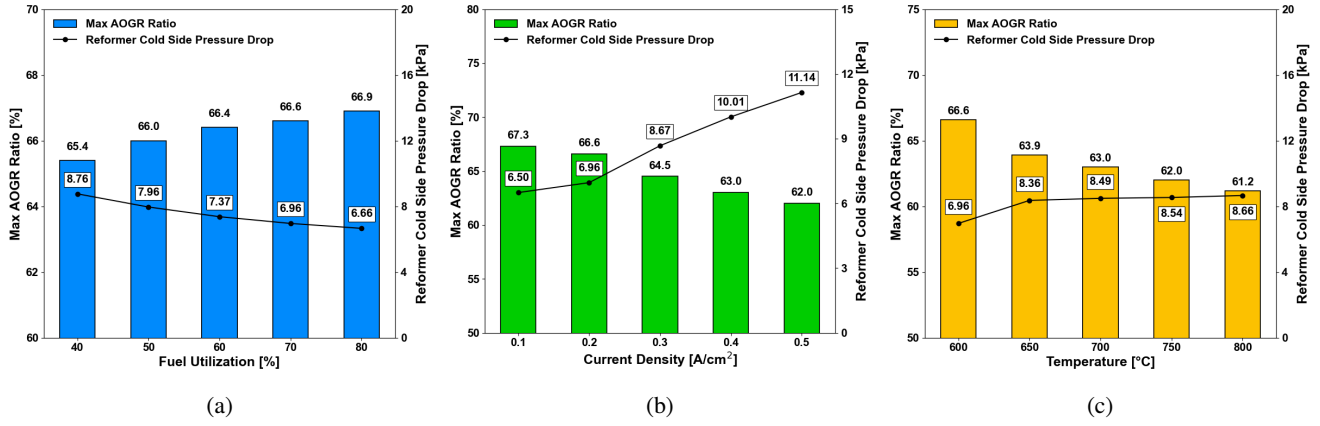


Figure 3: Maximum AOGR ratio permitted for variations in (a) FU_{stack} , (b) j , and (c) T_{stack} .

system efficiency from 33.9% to 51.3%, an absolute gain of 17.4 percentage points. At a high FU_{stack} of 80%, the same operation only improves the efficiency from 64.5% to 68.0%, yielding a much smaller gain of 3.5 percentage points. This asymmetry indicates that the efficiency benefit of AOGR is substantially greater at low FU_{stack} .

The origin of this asymmetry can be explained by the AOGR-induced reduction in the system fuel input. Under steady-state operation, the mass flow rate of ammonia required at the system inlet to sustain a target FU_{stack} is given by Eq. (7).

The first parenthetical term on the right-hand side represents the stoichiometric fuel consumption rate, while the second term, $(1 - r(1 - FU_{stack}))$, captures the effect of AOGR on the system-level fuel demand. Since $(1 - FU_{stack})$ is the unutilized fuel fraction exiting the stack, the term $r(1 - FU_{stack})$ quantifies the amount of unreacted fuel recovered through recirculation. At low FU_{stack} , a substantial fraction of unreacted fuel is available for recovery, so even modest AOGR ratios significantly reduce the fresh fuel demand. At high FU_{stack} , the recoverable amount is limited and AOGR provides only marginal savings. As shown in Fig. 4(b), at $FU_{stack} = 40\%$ the system fuel input expressed in terms of thermal energy decreases dramatically

from 157.6 kW to 95.7 kW, whereas at $FU_{stack} = 80\%$ it decreases only from 78.8 kW to 68.2 kW—a disparity that directly translates into the efficiency gain disparity observed in Fig. 4(a).

In contrast, the stack power density penalty associated with AOGR is nearly independent of FU_{stack} . As the AOGR ratio increases, the recirculated stream—rich in N_2 and H_2O —dilutes the fresh decomposed fuel and lowers the hydrogen partial pressure at the stack anode inlet. According to the Nernst equation, this reduction suppresses the reversible potential and consequently the cell voltage. As shown in Fig. 4(c), increasing the AOGR ratio from 0% to its maximum value reduces the cell voltage from 0.941 V to 0.878 V at $FU_{stack} = 40\%$ (a drop of 0.063 V), and from 0.903 V to 0.832 V at $FU_{stack} = 80\%$ (a drop of 0.071 V). The two voltage drops differ by less than 10 mV, indicating that the AOGR-induced voltage degradation is essentially comparable across the FU_{stack} range. This near-uniform behavior arises because the fuel dilution effect due to AOGR occurs in a similar magnitude among different FU_{stack} values. Consistent with this, the resulting power density reductions—from 0.188 W/cm² to 0.176 W/cm² at $FU = 40\%$ and from 0.181 W/cm² to 0.166 W/cm² at $FU = 80\%$ —differ by only 0.003 W/cm². Such results show that

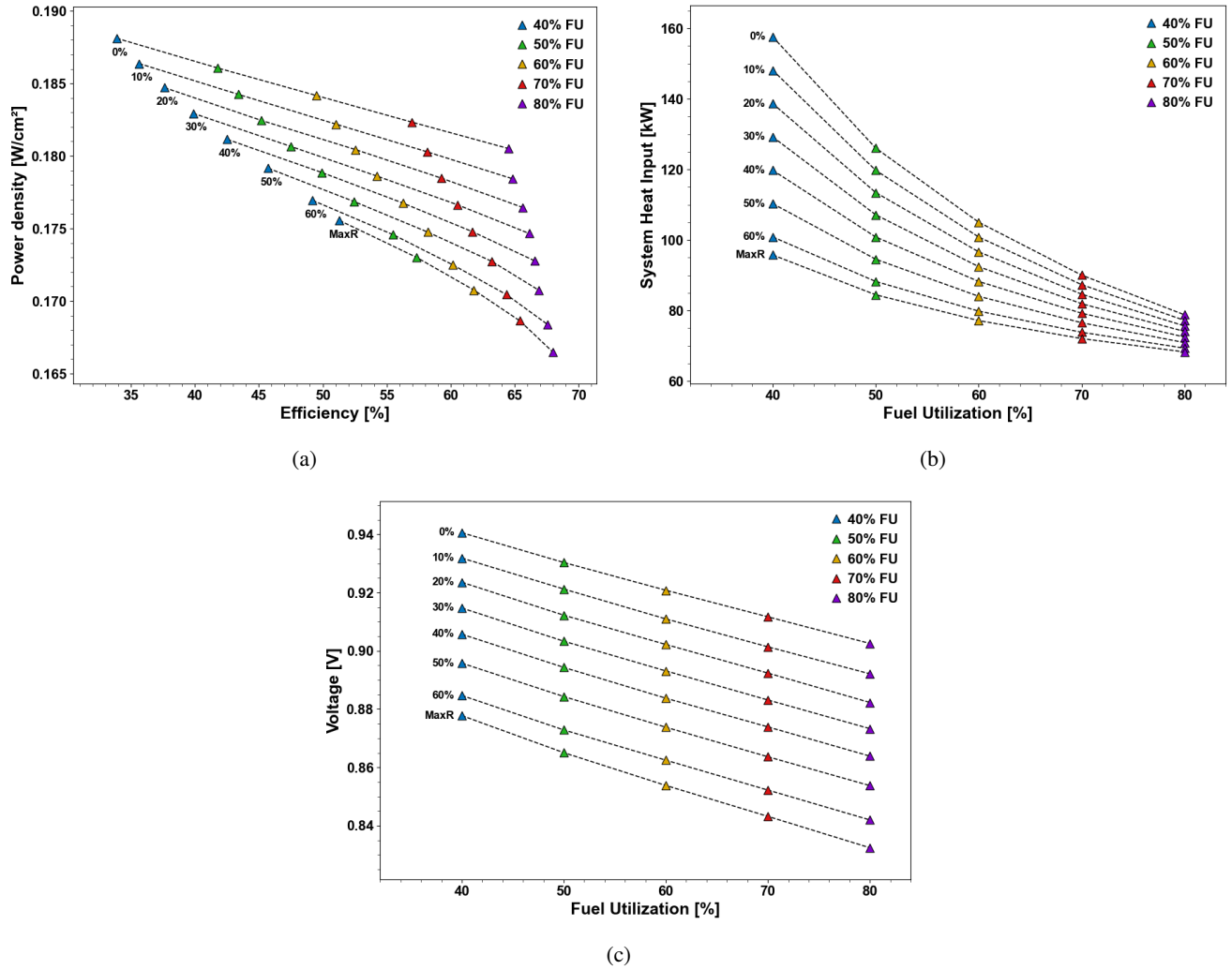


Figure 4: Performance map showing (a) efficiency and power density, (b) LHV, and (c) voltage as a function of FU_{stack} and AOGR ratio.

AOGR is most advantageous for SOFC systems operated at relatively low FU_{stack} .

4.3.2. Effect of AOGR ratio at different current density

System performance was next examined under varying j and AOGR ratio, and the resulting trajectories of system efficiency and stack power density are presented in Fig. 5(a). At a low j of 0.1 A/cm^2 , increasing the AOGR ratio from 0% to its maximum allowable value of 67.3% improves the system efficiency from 60.7% to 69.9%, corresponding to an absolute gain of 9.2 percentage points. At a high j of 0.5 A/cm^2 , however, the same operation only enhances the efficiency from 46.3% to 49.8%, yielding a much smaller

gain of 3.5 percentage points. Notably, the iso- j trajectories at high j exhibit a non-monotonic behavior: at 0.5 A/cm^2 , the system efficiency increases with the AOGR ratio up to approximately 60%, but then decreases as the AOGR ratio is further raised toward its maximum allowable value. The existence of an internal optimum at high j indicates that AOGR introduces a competing penalty that becomes increasingly dominant as j increases.

In contrast, the trend in stack power density with increasing AOGR ratio is far less sensitive to j . As shown in Fig. 5(b), the cell voltage at any given j decreases by a comparable magnitude as the AOGR ratio is increased,

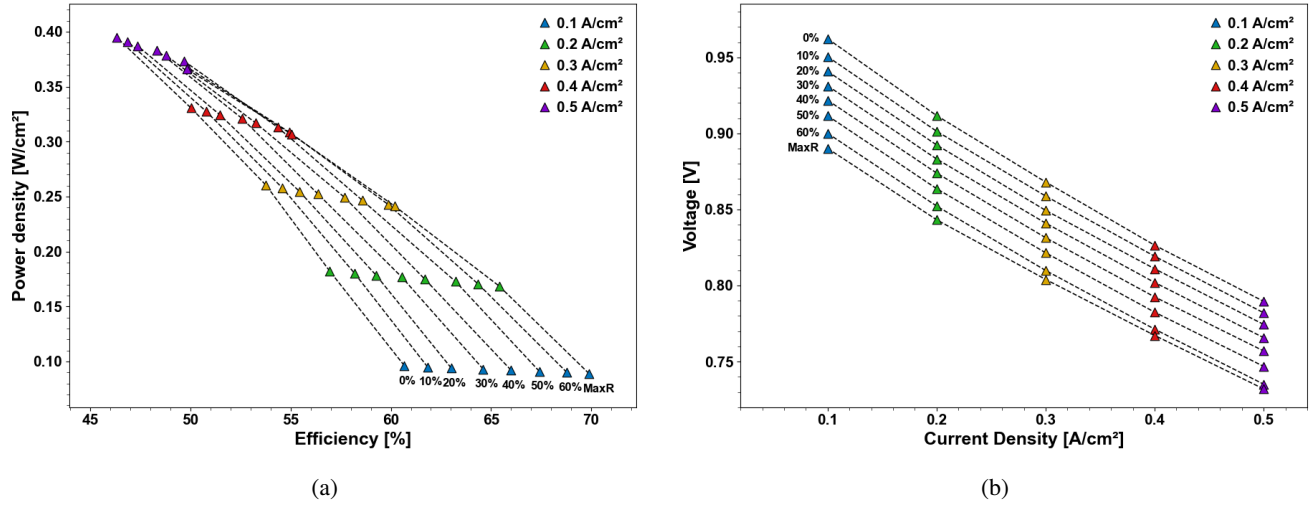


Figure 5: Performance map showing (a) efficiency and power density and (b) voltage as a function of j and AOGR ratio.

regardless of the absolute j level. Specifically, increasing the AOGR ratio from 0% to its maximum value reduces the cell voltage from 0.962 V to 0.890 V at 0.1 A/cm² (a drop of 0.072 V) and from 0.790 V to 0.732 V at 0.5 A/cm² (a drop of 0.058 V). The slightly larger voltage drop observed at low j is partly attributable to the significantly wider AOGR sweep range, since the maximum allowable AOGR ratio is 67.3% at 0.1 A/cm² but only 62.0% at 0.5 A/cm². When this difference in the AOGR sweep range is taken into account, the corresponding relative reductions in power density—from 0.096 W/cm² to 0.089 W/cm² at 0.1 A/cm² and from 0.395 W/cm² to 0.366 W/cm² at 0.5 A/cm²—correspond to nearly the same value of approximately 7% in both cases. This near-uniform power density degradation across j conditions can be understood from the fact that j does not significantly alter the molar composition at the stack inlet and outlet. Consequently, the resulting reduction in the anode hydrogen partial pressure is governed primarily by the AOGR ratio itself rather than by the operating j . Since the power density penalty alone cannot account for the non-monotonic efficiency behavior observed at high j , the source of the reversal must instead lie in the parasitic power consumption of the system BoP.

To clarify this, the system parasitic power was taken into analysis. As addressed in Eq. (10), the air blower is responsible for parasitic power consumption in the present system, and the pressure rise that the air blower must provide is determined by the cumulative pressure drop along the cathode-side flow path, given by Eq. (12).

$$p_{air\ blower} = 1\ atm + \Delta p_{stack} + \Delta p_{reformer,hot} + \Delta p_{air\ HX,hot} \quad (12)$$

where Δp_{stack} is the pressure drop across the SOFC stack cathode channel, $\Delta p_{reformer,hot}$ is the pressure drop across the hot side of the fuel reformer, and $\Delta p_{air\ HX,hot}$ is the pressure drop across the hot side of the air heat exchanger. The flow path of fluid to be pressurized by the air blower is represented in [Supplementary Fig. S2](#). As AOGR ratio increases, the pressure drops occurring at the heat exchanger and fuel reformer are amplified, while the SOFC stack channel pressure drop is determined almost constant as 0.5% of the fluid pressure. First, the pressure drop on the hot side of the fuel reformer rises through a more subtle mechanism than a simple increase in the cold-side flow rate. With increasing AOGR ratio, the mass flow rate of

fresh ammonia entering the reformer cold inlet decreases, since a larger fraction of the stack inlet fuel demand is met by recirculated unreacted fuel. This reduction in ammonia content moderately lowers the heat duty required for the endothermic NH_3 decomposition reaction. However, the AOGR ratio simultaneously raises the temperature of the reformer cold inlet stream, because the recirculated anode off-gas—returning at a high temperature—constitutes an increasingly large fraction of the cold stream. Since the reformer hot stream inlet temperature is essentially fixed as the stack cathode off gas temperature, the elevated cold-side temperature causes the log-mean temperature difference (LMTD) across the reformer to decrease substantially. Because the LMTD reduction outweighs the moderate decrease in the heat duty, the mass flow rate of the hot stream required to satisfy the heat balance must increase, leading to a higher pressure drop $\Delta p_{\text{reformer,hot}}$ on the hot side of the fuel reformer. The corresponding data of LMTD and heat duty at the fuel reformer is provided in [Supplementary Table S5](#) for different AOGR ratio operation. Second, the pressure drop on the hot side of the air heat exchanger rises through a chain of effects originating from the stack inlet-outlet temperature difference boundary condition. In SOFC operation, the cathode air flow rate is determined by the boundary condition that the temperature difference between the stack inlet and outlet must remain at 100°C . As discussed earlier, increasing the AOGR ratio reduces the cell operating voltage and consequently lowers the stack power generation. Since the total chemical energy released by the electrochemical reaction is partitioned between electrical work and heat, a reduction in the electrical power output translates directly into an increase in the heat that must be removed from the anode to maintain the prescribed inlet–outlet temperature difference. To dissipate this additional

heat, the cathode air flow rate must therefore be increased. This larger air flow rate elevates the heat capacity flow rate ($\dot{m}C_p$) on the cold side of the air heat exchanger, which receives the cathode air at ambient temperature and heats it up to T_{stack} . Since the cold stream now carries a greater thermal load to be transferred from the hot stream, the mass flow rate of the hot stream supplying the air heat exchanger must also increase to satisfy the heat balance. Consequently, $\Delta p_{\text{air HX,hot}}$ grows with the AOGR ratio. Combined with the direct increase in air flow rate through the blower itself, these compounding effects cause the air blower parasitic power to grow rapidly with the AOGR ratio.

The relative impact of this parasitic power growth on the overall system efficiency depends strongly on the operating j , since the absolute magnitude of both air flow rate and pressure drops occurring at the heat exchanger and fuel reformer scales with j . As summarized in [Table 5](#), at a low j of 0.1 A/cm^2 , the air blower parasitic power increases from 0.31 kW at zero recirculation to 0.52 kW at the maximum AOGR ratio.

The parasitic power increase is accompanied by an increase in $\Delta p_{\text{reformer,hot}}$ from 1.83 kPa to 3.20 kPa and in $\Delta p_{\text{air HX,hot}}$ from 1.92 kPa to 3.04 kPa. Although the absolute increase in parasitic power ($\approx 210 \text{ W}$) is non-negligible, it remains a minor fraction of the gross stack power output, corresponding to approximately 1% of the gross power generated under this operating condition. As a result, at low j the efficiency gain from reduced fuel input clearly dominates over the parasitic penalty, and the efficiency increases monotonically with the AOGR ratio. At a high j of 0.5 A/cm^2 , in contrast, the air blower parasitic power soars from 9.42 kW at zero recirculation to 14.13 kW at the maximum AOGR ratio—an absolute increase of approximately 22 times larger than the low- j case. The associated pressure drops on the

Table 5. Summary of parasitic power and pressure drops for different AOGR ratios at different j values

Parameters	Values										Units
j	0.1		0.2		0.3		0.4		0.5		A/cm ²
AOGR ratio	0	Max	0	Max	0	Max	0	Max	0	Max	%
$\Delta p_{\text{reformer, hot}}$	1.83	3.20	2.50	3.89	3.21	6.01	3.32	8.01	4.15	9.93	kPa
$\Delta p_{\text{air HX, hot}}$	1.92	3.04	3.49	3.69	5.23	6.69	8.32	10.37	12.04	14.68	kPa
$P_{\text{air blower}}$	0.31	0.52	1.23	1.44	2.37	3.88	5.11	7.99	9.42	14.13	kW

hot side of the fuel reformer and the air heat exchanger also rise much more steeply, from 4.15 kPa to 9.93 kPa and from 12.04 kPa to 14.68 kPa, respectively. The increase in parasitic power at this condition reaches approximately 4.5% of the gross stack power output at the maximum AOGR ratio, which is significant enough to overcome the benefit of reduced fuel input beyond a certain AOGR ratio. Consequently, at 0.5 A/cm², the system efficiency continues to improve only up to about 60% AOGR; beyond this point, the marginal increase in parasitic power exceeds the marginal reduction in fuel input, and the efficiency begins to decline despite the continued recovery of unutilized fuel. As a result, the dominant trade-off under varying j shifts from the stack level to the balance-of-plant level, such that the optimal AOGR ratio at high j lies below the maximum allowable value and decreases further as j increases.

4.3.3. Effect of AOGR ratio at different stack inlet temperatures

The effect of AOGR ratio on the system performance was finally examined under varying T_{stack} conditions, ranging from 600 °C to 800 °C. The resulting trajectories of system efficiency and stack power density are presented in Fig. 6(a). At zero recirculation, the highest system performance is achieved at a T_{stack} of 750 °C, with the operating point at 700 °C closely following and effectively overlapping with the

750 °C case in the efficiency–power density plane. Specifically, the corresponding power densities and system efficiencies are virtually identical at 0.1908 W/cm² and 59.92%, respectively. The 600 °C and 650 °C operating points yield progressively lower performance than the 700–750 °C cases, while the 800 °C case—despite being the highest temperature considered—delivers performance that is consistently lower than that of the 700–750 °C cases throughout the entire AOGR range. This indicates that the cell voltage at zero recirculation already peaks within the 700–750 °C interval and operating at 800 °C lies on the descending side of the voltage–temperature curve. As the AOGR ratio increases, the trajectories of the higher-temperature cases (750 °C and 800 °C) curve downward more steeply than that of the 700 °C case, and the relative ranking among the operating temperatures evolves accordingly. At maximum allowable AOGR ratio, 700 °C emerges as the most efficient operating temperature with a system efficiency of 67.52%, while 750 °C falls slightly behind at 67.02% and 800 °C reaches only 66.02%.

Such result in system performance originates from the effect of AOGR ratio and T_{stack} on stack voltage. As shown in Fig. 6(b), the cell voltage at any given AOGR ratio exhibits a non-monotonic dependence on T_{stack} : it first increases from 600 °C to an intermediate maximum and then decreases again toward 800 °C. An intermediate optimal temperature exists due to a trade-off in reversible potential

and overpotential. As the operating temperature of the SOFC stack increases, its activation and ohmic overpotential decreases due to a decrease in exchange current density and increase in ionic conductivity, while the reversible potential also decreases. Detailed information in stack voltage at different T_{stack} with base case FU_{stack}, j , AOGR ratio, is provided in [Supplementary Table S6](#). The location of this maximum, however, shifts to lower temperatures as the AOGR ratio increases. At zero recirculation, the cell voltage peaks broadly across the 700–750 °C range, with the two temperatures yielding nearly identical voltage values of 0.9541 V and 0.9542 V. As the AOGR ratio increases, the peak gradually narrows and migrates toward 700 °C: at the maximum allowable AOGR ratio, the cell voltage at 700 °C (0.8807 V) clearly exceeds that at 750 °C (0.8779 V), even though the two values were essentially indistinguishable at zero recirculation. Reflecting this temperature-dependent penalty, the cumulative voltage drops incurred over the full AOGR sweep increases monotonically with T_{stack} , from 68.5 mV at 600 °C to 73.3 mV at 700 °C, 76.3 mV at 750 °C, and 79.1 mV at 800 °C, even though the maximum available AOGR ratio is lower at a high T_{stack} . This trend confirms that AOGR introduces a temperature-dependent penalty mechanism whose magnitude grows with the AOGR ratio and disproportionately affects the higher-temperature operating points.

The origin of this behavior can be explained by the dependence of reversible potential—which is the dominant factor determining cell voltage—on temperature and anode gas compositions, as higher temperatures reduce the reversible potential. The reversible cell voltage of the SOFC is given by Eq. (2) where the second term on the right-hand side—the Nernstian term—is responsible for the temperature dependence of the reversible potential under the

operating conditions of interest. As the AOGR ratio increases, the recirculated stream depletes hydrogen partial pressure p_{H_2} and enriches the steam partial pressure p_{H_2O} at the stack anode inlet, which reduces the partial pressure ratio inside the logarithm to a value below unity and renders the logarithmic argument increasingly negative in sign. Therefore, at higher AOGR ratios, a given temperature increment produces a larger reduction in the Nernstian term and therefore a steeper decline in the reversible potential. To quantify the amplification of the reversible potential-reduction effect with AOGR, the magnitude of the reversible potential temperature sensitivity, $|dE_{rev}/dT|$, is shown in [Supplementary Fig. S3](#). This finding suggests that the design and operation of NH_3 -fueled SOFC systems incorporating AOGR should explicitly account for the coupling between the AOGR ratio and T_{stack} , rather than treating the two parameters as independently optimizable variables. In particular, when high AOGR ratios are desired for system efficiency reasons, operating the stack at moderately high temperatures yields superior overall performance compared to operating at the upper end of the temperature range.

5. Conclusions

A comprehensive parametric investigation of an ammonia-fueled SOFC system incorporating anode off-gas recirculation (AOGR) was conducted to clarify both the operating limits imposed on the AOGR ratio and the influence of the AOGR ratio on the overall system performance. An integrated thermodynamic system model based on validated component models—including a 1-D SOFC stack, shell-and-tube heat exchanger, packed-bed ammonia reformer, and critical-mode ejector—was constructed, and three operating parameters (FU_{stack}, j , and T_{stack}) were systematically

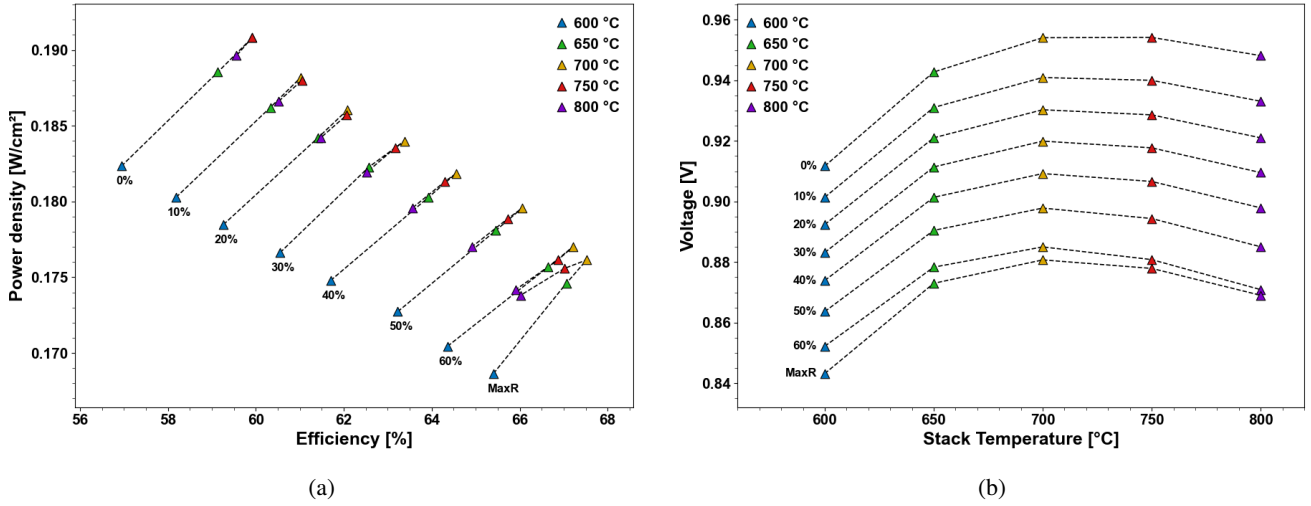


Figure 6: Performance map showing (a) efficiency and power density and (b) voltage as a function of T_{stack} and AOGR ratio.

varied together with the AOGR ratio. The principal findings are summarized as follows.

- The maximum allowable AOGR ratio is governed by the pressure-balance constraint of the AOGR loop, in which the ejector pressure lift must overcome the cumulative pressure drops along the recirculation path. Increasing FU_{stack} expands this limit modestly by reducing the reformer cold-side pressure drop, whereas increasing j narrows it substantially due to the larger anode-side flow rate and the longer reformer length required to meet the elevated heat-duty demand.
- T_{stack} limits the maximum allowable AOGR ratio through a fundamentally distinct mechanism. The elevated entrained flow temperature degrades the entrainment and pressure lift performance of the ejector, thereby reducing the maximum allowable AOGR ratio more sharply than what the pressure drop alone would predict.
- Under varying FU_{stack} , the AOGR ratio induces a strongly asymmetric trade-off: the efficiency gain is highly sensitive to FU_{stack} , whereas the power density penalty remains nearly insensitive to it. AOGR is

therefore most advantageous at low FU_{stack} , where a substantial efficiency improvement is achieved at the same power density cost as at high FU_{stack} .

- Under varying j , the dominant trade-off shifts from the stack level to the balance-of-plant level, giving rise to a non-monotonic efficiency trajectory at high j . The air blower parasitic power, amplified by the elevated air flow rate and the increased BoP pressure drops, eventually overcomes the benefit of reduced fuel input. Consequently, the optimal AOGR ratio lies below the maximum allowable value identified from the pressure-balance constraint.
- Under varying T_{stack} , the optimal operating temperature shifts downward as the AOGR ratio increases, since the AOGR-induced enrichment of steam at the anode inlet amplifies the reversible potential's temperature sensitivity. The resulting steeper reversible potential decline at higher temperatures overwhelms the modest overpotential reduction available under low-current-density operation, so that operation at the upper end of the investigated temperature range no

longer yields the best performance under high AOGR conditions.

These findings demonstrate that the AOGR ratio cannot be treated as an independently optimizable variable in NH_3 -fueled SOFC systems, since both its operating limits and its influence on system performance are tightly coupled to the underlying operating conditions through distinct physical mechanisms. The performance maps derived in this work provide quantitative guidance for selecting the AOGR ratio in conjunction with each operating parameter and offer practical design insights for NH_3 -fueled SOFC systems targeting marine power generation applications.

Despite the comprehensive scope of the present analysis, several limitations should be acknowledged and addressed in future work. First, the geometrical design variables of the BoP components—including the shell-and-tube heat exchanger and the packed-bed ammonia reformer—were not individually optimized for each operating condition examined in this study, although their dimensions were resized when the heat-duty requirements could not be satisfied. A condition-specific optimization of these geometrical variables could further refine the trade-offs identified in this work and yield more accurate quantification of the AOGR-induced parasitic power growth. Second, the present analysis was based on a single fixed system layout—the modified Ejector CG configuration—and alternative system architectures such as blower-driven recirculation, hybrid ejector-blower configurations, or layouts with different heat integration schemes were not considered, even though such variations may shift the location of the dominant trade-off mechanism and the resulting optimal AOGR ratio. Third, stack durability and material degradation under prolonged AOGR operation were not included in the present scope, although the AOGR-induced enrichment of steam at the anode inlet

identified in this work suggests that long-term durability considerations could further constrain the practical AOGR operating window beyond the steady-state thermodynamic limits established here. Future work should therefore extend the present framework by performing condition-specific optimization of the BoP geometries, comparatively evaluating alternative system layouts, and integrating stack durability models to translate the system-level design insights of this study into long-term operating guidelines for NH_3 -fueled SOFC systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

To be completed by the authors before submission.

References

- [1] I. Robalo-Cabrera, A. Alcayde, A. Filgueira-Vizoso, L. Castro-Santos, A. García-Diez, F. Manzano-Agugliaro, [Shipping sector decarbonisation measures: A review](https://doi.org/10.1016/j.seta.2025.104549), *Sustainable Energy Technologies and Assessments* 82 (2025) 104549. doi:<https://doi.org/10.1016/j.seta.2025.104549>. URL <https://www.sciencedirect.com/science/article/pii/S2213138825003807>
- [2] International Maritime Organization, 2023 *imo strategy on reduction of ghg emissions from ships*, <https://www.imo.org/en/ourwork/environment/pages/>

- 2023-imo-strategy-on-reduction-of-ghg-emissions-from-shipping, accessed: 2026-07-07 (2023).
- [3] A. Al-Enazi, E. C. Okonkwo, Y. Bicer, T. Al-Ansari, *A review of cleaner alternative fuels for maritime transportation*, *Energy Reports* 7 (2021) 1962–1985. doi:<https://doi.org/10.1016/j.egy.2021.03.036>. URL <https://www.sciencedirect.com/science/article/pii/S2352484721002067>
- [4] G. Voniati, A. Dimaratos, G. Koltsakis, L. Ntzichristos, *Ammonia as a marine fuel towards decarbonization: Emission control challenges*, *Sustainability* 15 (21) (2023). doi:[10.3390/su152115565](https://doi.org/10.3390/su152115565). URL <https://www.mdpi.com/2071-1050/15/21/15565>
- [5] M. Aziz, A. T. Wijayanta, A. B. D. Nandiyanto, *Ammonia as effective hydrogen storage: A review on production, storage and utilization*, *Energies* 13 (12) (2020). doi:[10.3390/en13123062](https://doi.org/10.3390/en13123062). URL <https://www.mdpi.com/1996-1073/13/12/3062>
- [6] C. J. McKinlay, S. R. Turnock, D. A. Hudson, *Route to zero emission shipping: Hydrogen, ammonia or methanol?*, *International Journal of Hydrogen Energy* 46 (55) (2021) 28282–28297. doi:<https://doi.org/10.1016/j.ijhydene.2021.06.066>. URL <https://www.sciencedirect.com/science/article/pii/S0360319921022175>
- [7] Alfa Laval, Hafnia, Haldor Topsøe, Vestas, Siemens Gamesa, *Ammonfuel: An industrial view of ammonia as a marine fuel*, accessed: 2026-07-07 (Aug. 2020). URL <https://www.alfalaval.com/globalassets/documents/industries/marine-and-transportation/marine/fcm-lff/ammonia-as-fuel/ammonfuel-report-version-09.9-august-3.pdf>
- [8] M. Nygård, J. Wärnå, V. Russo, J. Torrkulla, D. Y. Murzin, *Kinetic modelling of catalytic N_2O removal*, *Scientific Reports* 15 (2025) 44959. doi:[10.1038/s41598-025-28647-6](https://doi.org/10.1038/s41598-025-28647-6). URL <https://doi.org/10.1038/s41598-025-28647-6>
- [9] L. van Biert, M. Godjevac, K. Visser, P. Aravind, *A review of fuel cell systems for maritime applications*, *Journal of Power Sources* 327 (2016) 345–364. doi:<https://doi.org/10.1016/j.jpowsour.2016.07.007>. URL <https://www.sciencedirect.com/science/article/pii/S0378775316308631>
- [10] D. S. Dhawale, S. Biswas, G. Kaur, S. Giddey, *Challenges and advancement in direct ammonia solid oxide fuel cells: a review*, *Inorganic Chemistry Frontiers* 10 (21) (2023) 6176–6192. arXiv:<https://pubs.rsc.org/qi/article-pdf/10/21/6176/9293816/d3qi01557b.pdf>, doi:[10.1039/d3qi01557b](https://doi.org/10.1039/d3qi01557b). URL <https://doi.org/10.1039/d3qi01557b>
- [11] G. Jeerh, M. Zhang, S. Tao, *Recent progress in ammonia fuel cells and their potential applications*, *Journal of Materials Chemistry A* 9 (2) (2021) 727–752. arXiv:<https://pubs.rsc.org/ta/article-pdf/9/2/727/7617518/d0ta08810b.pdf>, doi:[10.1039/d0ta08810b](https://doi.org/10.1039/d0ta08810b). URL <https://doi.org/10.1039/d0ta08810b>
- [12] A. Wojcik, H. Middleton, I. Damopoulos, J. Van herle, *Ammonia as a fuel in solid oxide fuel cells*, *Journal of Power Sources* 118 (1) (2003) 342–348, *scientific Advances in Fuel Cell Systems*. doi:[https://doi.org/10.1016/S0378-7753\(03\)00083-1](https://doi.org/10.1016/S0378-7753(03)00083-1). URL <https://www.sciencedirect.com/science/article/pii/S0378775303000831>
- [13] G. Cinti, G. Discepoli, E. Sisani, U. Desideri, *Sofc operating with ammonia: Stack test*

- and system analysis, *International Journal of Hydrogen Energy* 41 (31) (2016) 13583–13590.
doi:<https://doi.org/10.1016/j.ijhydene.2016.06.070>.
URL <https://www.sciencedirect.com/science/article/pii/S036031991630595X>
- [14] F. Ishak, I. Dincer, C. Zamfirescu, *Thermodynamic analysis of ammonia-fed solid oxide fuel cells*, *Journal of Power Sources* 202 (2012) 157–165.
doi:<https://doi.org/10.1016/j.jpowsour.2011.10.142>.
URL <https://www.sciencedirect.com/science/article/pii/S0378775311022774>
- [15] J. Lee, K. Jeong, Y. Choi, Y. J. Park, J. Hong, *Thermodynamic analysis of internal and external ammonia decomposition strategies for ammonia-fueled solid oxide fuel cell systems*, *Journal of Power Sources* 658 (2025) 238271.
doi:<https://doi.org/10.1016/j.jpowsour.2025.238271>.
URL <https://www.sciencedirect.com/science/article/pii/S037877532502107X>
- [16] F. Ishak, I. Dincer, C. Zamfirescu, *Energy and exergy analyses of direct ammonia solid oxide fuel cell integrated with gas turbine power cycle*, *Journal of Power Sources* 212 (2012) 73–85.
doi:<https://doi.org/10.1016/j.jpowsour.2012.03.083>.
URL <https://www.sciencedirect.com/science/article/pii/S0378775312007045>
- [17] R. Torii, Y. Tachikawa, K. Sasaki, K. Ito, *Anode gas recirculation for improving the performance and cost of a 5-kw solid oxide fuel cell system*, *Journal of Power Sources* 325 (2016) 229–237.
doi:<https://doi.org/10.1016/j.jpowsour.2016.06.045>.
URL <https://www.sciencedirect.com/science/article/pii/S0378775316307479>
- [18] K. Selvam, Y. Komatsu, A. Sciazko, S. Kaneko, N. Shikazono, *Thermodynamic analysis of 100% system fuel utilization solid oxide fuel cell (sofc) system fueled with ammonia*, *Energy Conversion and Management* 249 (2021) 114839.
doi:<https://doi.org/10.1016/j.enconman.2021.114839>.
URL <https://www.sciencedirect.com/science/article/pii/S0196890421010153>
- [19] K. Jeong, J. Lee, J. Hong, *Comparative performance analysis of hydrogen-, methane-, and ammonia-fueled solid oxide fuel cell systems considering anode off-gas recirculation*, *Energy Conversion and Management: X* 30 (2026) 101799.
doi:<https://doi.org/10.1016/j.ecmx.2026.101799>.
URL <https://www.sciencedirect.com/science/article/pii/S2590174526002825>
- [20] M. Bakała, K. Martsinchyk, K. Motyliński, J. Kupecki, K. Machaj, *Numerical analysis of natural gas, hydrogen and ammonia fueled solid oxide fuel cell based micro cogeneration units with anodic gas recirculation*, *International Journal of Hydrogen Energy* 52 (2024) 952–964.
doi:<https://doi.org/10.1016/j.ijhydene.2023.06.267>.
URL <https://www.sciencedirect.com/science/article/pii/S0360319923032469>
- [21] T.-Q. Quach, V.-T. Giap, D. Keun Lee, T. Pineda Israel, K. Young Ahn, *High-efficiency ammonia-fed solid oxide fuel cell systems for distributed power generation*, *Applied Energy* 324 (2022) 119718.
doi:<https://doi.org/10.1016/j.apenergy.2022.119718>.
URL <https://www.sciencedirect.com/science/article/pii/S0306261922010091>

- [22] B. Sun, Y. Xu, Y. Liu, Y. Ya, X. Cheng, Y. Quan, H. Ding, J. Xie, Numerical design and evaluation of ammonia fueled solid oxide fuel cell (sofc) power systems with different integration architecture: Efficiency, power density and thermal safety, Energy Conversion and Management 298 (2023) 117804. doi:<https://doi.org/10.1016/j.enconman.2023.117804>. URL <https://www.sciencedirect.com/science/article/pii/S0196890423011500>
- [23] S. Oh, S. Kim, J. Jung, J. S. Ahn, S. Kang, Numerical analysis and parametric optimization of a direct ammonia solid oxide fuel cell system integrated with organic rankine cycle, Journal of Power Sources 620 (2024) 235241. doi:<https://doi.org/10.1016/j.jpowsour.2024.235241>. URL <https://www.sciencedirect.com/science/article/pii/S0378775324011935>
- [24] IAPWS Industrial Formulation 1997 for the Thermodynamic Properties of Water and Steam, Springer Berlin Heidelberg, Berlin, Heidelberg, 2008, pp. 7–150. doi:[10.1007/978-3-540-74234-0_3](https://doi.org/10.1007/978-3-540-74234-0_3). URL https://doi.org/10.1007/978-3-540-74234-0_3
- [25] G. P. Smith, D. M. Golden, M. Frenklach, N. W. Morarty, B. Eiteneer, M. Goldenberg, C. T. Bowman, R. K. Hanson, S. Song, W. C. Gardiner, V. V. Lissianski, Z. Qin, GRI-Mech 3.0, http://www.me.berkeley.edu/gri_mech/, accessed: 2026-07-09 (1999).
- [26] B. Todd, J. Young, Thermodynamic and transport properties of gases for use in solid oxide fuel cell modelling, Journal of Power Sources 110 (1) (2002) 186–200. doi:[https://doi.org/10.1016/S0378-7753\(02\)00277-X](https://doi.org/10.1016/S0378-7753(02)00277-X). URL <https://www.sciencedirect.com/science/article/pii/S037877530200277X>
- [27] M. J. Moran, H. N. Shapiro, D. D. Boettner, M. B. Bailey, Moran's principles of engineering thermodynamics, (No Title) (2017).
- [28] J. Lee, Thermodynamic design and analysis of ammonia-fueled solid oxide fuel cell system depending on ammonia decomposition types (2023).
- [29] Y. J. Park, G. Min, J. Hong, Comparative study of solid oxide fuel cell-combined heat and power system designs for optimal thermal integration, Energy Conversion and Management 182 (2019) 351–368. doi:<https://doi.org/10.1016/j.enconman.2018.12.068>. URL <https://www.sciencedirect.com/science/article/pii/S0196890418314134>
- [30] A. Fuerte, R. Valenzuela, M. Escudero, L. Daza, Ammonia as efficient fuel for sofc, Journal of Power Sources 192 (1) (2009) 170–174, cONAPPICE 2008, Zaragoza, Spain, 24-26 September 2008. doi:<https://doi.org/10.1016/j.jpowsour.2008.11.037>. URL <https://www.sciencedirect.com/science/article/pii/S0378775308021873>
- [31] I. Lucentini, A. Casanovas, J. Llorca, Catalytic ammonia decomposition for hydrogen production on ni, ru and niru supported on ceo2, International Journal of Hydrogen Energy 44 (25) (2019) 12693–12707, special Issue on Selected Contributions from the European Hydrogen Energy Conference 2018. Málaga, Spain. March 14th - 16th. doi:<https://doi.org/10.1016/j.ijhydene.2019.01.154>. URL <https://www.sciencedirect.com/science/article/pii/S0360319919302861>

- [32] N. Maffei, L. Pelletier, A. McFarlan, A high performance direct ammonia fuel cell using a mixed ionic and electronic conducting anode, *Journal of Power Sources* 175 (1) (2008) 221–225. doi:<https://doi.org/10.1016/j.jpowsour.2007.09.040>. URL <https://www.sciencedirect.com/science/article/pii/S0378775307019933>
- [33] Y. J. Park, G. Min, J. Hong, Optimal method for the anode exhaust gas recycling of atmospheric solid oxide fuel cell-combined heat and power systems, *Journal of Power Sources* 567 (2023) 232970. doi:<https://doi.org/10.1016/j.jpowsour.2023.232970>. URL <https://www.sciencedirect.com/science/article/pii/S0378775323003452>
- [34] F. Marsano, L. Magistri, A. Massardo, Ejector performance influence on a solid oxide fuel cell anodic recirculation system, *Journal of Power Sources* 129 (2) (2004) 216–228. doi:<https://doi.org/10.1016/j.jpowsour.2003.11.034>. URL <https://www.sciencedirect.com/science/article/pii/S037877530301125X>
- [35] E. Ozden, I. Tari, Shell side cfd analysis of a small shell-and-tube heat exchanger, *Energy Conversion and Management* 51 (5) (2010) 1004–1014. doi:<https://doi.org/10.1016/j.enconman.2009.12.003>. URL <https://www.sciencedirect.com/science/article/pii/S0196890409005020>
- [36] DNV, Ammonia as a Marine Fuel Safety Handbook, Tech. Rep. Rev. 02, Green Shipping Programme, developed by DNV on behalf of the Green Shipping Programme, with input from the Norwegian Maritime Authority (Jun. 2023).
- [37] S. Seidler, M. Henke, J. Kallo, W. G. Bessler, U. Maier, K. A. Friedrich, Pressurized solid oxide fuel cells: Experimental studies and modeling, *Journal of Power Sources* 196 (17) (2011) 7195–7202, proceedings of 2010 European Solid Oxide Fuel Cell Forum. doi:<https://doi.org/10.1016/j.jpowsour.2010.09.100>. URL <https://www.sciencedirect.com/science/article/pii/S0378775310017076>
- [38] G. Cinti, V. Liso, S. S. Araya, Design improvements for ammonia-fed sofc systems through power rating, cascade design and fuel recirculation, *International Journal of Hydrogen Energy* 48 (40) (2023) 15269–15279. doi:<https://doi.org/10.1016/j.ijhydene.2023.01.008>. URL <https://www.sciencedirect.com/science/article/pii/S0360319923000113>
- [39] K. Thulukkanam, Heat exchanger design handbook, CRC press, 2000.
- [40] S. Kakaç, H. Liu, A. Pramuanjaroenkij, Heat exchangers: selection, rating, and thermal design, CRC press, 2002.
- [41] J. R. Thome, Wolverine engineering databook iii (2006).
- [42] J. E. Edwards, Design and rating shell and tube heat exchangers, Teesside, Ukraine (2008).
- [43] L. Wang, C. Wang, W. Hou, H. Zhao, H. Zhang, Experimental investigation on ejector performance near critical back pressure, *International Journal of Refrigeration* 80 (2017) 158–168. doi:<https://doi.org/10.1016/j.ijrefrig.2017.05.010>. URL <https://www.sciencedirect.com/science/article/pii/S0140700717301998>

- [44] T. Liang, Y. Wang, D. Zhang, Z. Li, W. Ye, G. Li, W. Xu, Numerical investigation on the mixing process within the two-strut supersonic ejector, International Journal of Heat and Fluid Flow 116 (2025) 109984. doi:<https://doi.org/10.1016/j.ijheatfluidflow.2025.109984>.
URL <https://www.sciencedirect.com/science/article/pii/S0142727X25002425>